

Detecting Anomalous Links in Networked Systems Exhibiting Scale-Free-Like Behaviors

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Abstract—Anomaly detection acts as the first line of defense against cyber-attacks to ensure the security and integrity of networked systems. Recent anomaly detection techniques employ dynamic graph representations to capture the complex interactions in networked systems that evolve over time. However, many existing methods focus on identifying anomalous node behavior and often rely on additional information, such as node features, node attributes, and edge features to detect them effectively. In this study, we propose a probabilistic approach to predict future anomalous links in dynamically evolving scale-free-like networks. Specifically, we introduce three probabilistic models exploiting the scale-free property of networked systems based on past communication patterns. We perform a grid search on the receiver operating characteristic curves of the proposed models to determine the best decision boundaries for identifying anomalous links. Our approach is able to determine the direction of future links and estimate their probabilities based on link orientations. We evaluate our proposed models on a synthetic dataset derived from a dynamically evolving real-world communication network. Our empirical results show that the accuracy rates of the proposed models change between 63.20% to 96.50%, while the F1 score is between 66.25% and 96.52%. Lastly, we show that the proposed approach outperforms two popular walk-based and two contemporary neural network-based methods.

Index Terms—Anomalous link detection, complex networks, dynamic graphs, link prediction, network security.

I. INTRODUCTION

DEVICES in a networked system communicate with each other to exchange information and share resources through a communication medium [1]. The medium of communication can be wired, wireless, or a combination of them. A networked system may contain a variety of devices, such as computers, networking equipment, smart home appliances, wearables, and Internet of Things (IoT) devices. Networked systems can also

be categorized into different types, including personal area networks consisting of personal devices communicating with each other in a close proximity [2], home area networks embracing IoT devices that enable home automation and control within a household [3], local area networks composed of computers and peripherals communicating within a confined area [4], metropolitan area networks encompassing surveillance devices and networking equipment communicating within a city's boundaries [3], and wide area networks made up of equipment that communicate across a vast geographical region to support a cellular telecommunications infrastructure [5].

In such networked systems, the devices or hosts frequently undertake diverse roles. For instance, sensors gather and transmit data, routers manage network traffic, computing devices handle information processing tasks, servers provide digital services, client devices consume digital services, and peer devices assume both roles interchangeably. Although the hosts may fulfill various roles, these roles typically remain stable over time. Hence, networked systems demonstrate consistent communication behavior between successive time intervals [6].

Graphs are commonly employed to represent and analyze networked systems. Each vertex or node in the graph refers to a device in the network, and each edge or link denotes the communication between two devices. Previous studies have established that many networked systems exhibit the scale-free property. That is, the degree of a vertex follows a power-law distribution, expressed as $P(k) \sim k^{-\gamma}$, where $2 < \gamma < 3$ [7]. However, recent research indicates that true scale-free networks are not that common, and their definitions vary across published literature [8]. According to [8], the definitions of scale-free property range from “super-weak” scale-free to “strongest” scale-free. In this study, we focus on the networks demonstrating “scale-free-like” behavior, which entails 1) the majority of vertices have lower degrees and fewer vertices have higher degrees, 2) vertices with higher degrees are not rare, indicating that their degree distributions have heavier tails [9].

Networked systems are typically designed with a client-server model in consideration, rather than a peer-to-peer model. Therefore, such systems often demonstrate scale-free-like behavior. That is, there exists a greater number of service consumers (clients) than service providers (servers) within a network [10]. Although some servers may receive ancillary services from other servers, these systems often establish a network consisting of hub vertices (servers) and peripheral vertices (clients). Besides, the roles of the vertices remain stable over time. Hence, such networked systems demonstrate a consistent pattern in the communication behaviors between successive time intervals.

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In this study, we leverage the scale-free-like property of networked systems to identify anomalous communication patterns that may pose a threat to the systems. Anomalous communication inceptions in real-world networked systems may indicate the following:

- 1) a worm attempting to propagate from a server to client machines that have previously received a service;
- 2) a malevolent actor periodically scanning computers for known or unknown vulnerabilities;
- 3) an employee attempting to gain access to computers that he/she has not previously interacted with.

Identifying anomalous links has applications across diverse domains. Examples include the following:

- 1) a surge of interactions with the accounts of prominent or public figures in social networks;
- 2) an abundance of links to seasonal items in recommendation networks;
- 3) connections between proteins or genes that are not traditionally recognized to interact in protein-protein networks.

Nonetheless, our research specifically focuses on the scale-free-like networked systems and their graph representations that dynamically evolve over time.

Several studies focus on anomaly detection in dynamic graphs [11], [12], [13], [14], [15], [16], [17]. A significant portion of these studies concentrate on identifying anomalous nodes and/or are limited to undirected graphs. In addition, existing studies require additional information such as node features, node attributes, or edge features [12], [13], [18]. In this study, we propose three probabilistic models to predict future anomalous links in dynamically evolving scale-free-like networks. Our proposed models exploit the degrees of the nodes in a system to identify anomalous links. Lower link probabilities indicate anomalous communication inceptions, whereas higher link probabilities imply expected communications. Hence, the proposed models can be employed for both anomalous link detection and expected link prediction tasks. However, the primary goal of this study is to detect anomalous links. We explicitly define these models over undirected and directed graphs. To learn and assign a probability to each node pair, we employ a randomly selected training dataset that consists of different types of node pairs. Next, we perform a grid search on the receiver operating characteristic (ROC) curve of the training dataset to find an optimal decision boundary to classify future links as either anomalous or expected. Lastly, we introduce an algorithm leveraging the decision boundary and domain expertise to automate or semiautomate future link acceptances and rejections in a network.

Experimental results indicate that our proposed approach attains high accuracy, precision, and recall rates for both anomalous link detection and expected link prediction tasks. We evaluate the proposed models on a synthetic dataset provided by the MITRE Corporation. MITRE employed an AI model that decomposes synthetic iterations of a real-world network communication dataset while preserving its authentic characteristics. Our analysis in Section IV-D shows that the network snapshots adhere to the “weak” and “weakest” designations of the scale-free networks [8]. Our empirical evaluations demonstrate that the proposed model reach up to an average accuracy of 96.50%, average precision of 95.87%, and average recall of 97.19% over

directed graphs. In addition, we develop a *worst-case scenario* to test the robustness of our proposed approach, where an anomalous link may appear in the opposite direction of an expected link. Our proposed model defined over directed graphs can successfully detect such anomalous links with an average accuracy of 96.66%. Furthermore, the comparative analyses show that our approach consistently outperforms two popular walk-based methods, i.e., Node2Vec [19] and DeepWalk [20], and two contemporary neural network-based methods, i.e., TADDY [21] and DGCN [22] over directed graphs.

The main contributions of this study are three-fold.

- 1) We propose three probabilistic models defined over undirected and directed graphs to identify anomalous communication links in evolving networked systems.
- 2) Our proposed models exploit the scale-free-like property of networked systems to successfully classify future anomalous and expected links.
- 3) We present a *worst-case scenario* to evaluate the robustness of our proposed models in identifying one-way-missing anomalous links.

We developed the proposed models for the networked systems that exhibit scale-free-like behavior. Hence, it is essential to test the scale-freeness of a networked system before applying the models, as proposed in [8].

Note that a preliminary version of this article appeared in the proceedings of the IEEE International Systems Conference, 2024 [23]. In this archival version, we introduce the theoretical proofs of the probabilistic models, present new experiments conducted on the *worst-case scenario*, implement two popular walk-based and two contemporary neural network-based methods to compare with our approach, provide a more critical analysis of existing methods and extend the evaluations by discussing the latest experimental results and demonstrating a comprehensive comparative analyses.

The rest of this article is organized as follows. Section II presents the related work. In Section III, we explain the probabilistic models proposed for anomalous link prediction. Section IV describes the dataset and discusses the empirical evaluations of our approach. In Section V, we compare the proposed approach with two popular walk-based and two contemporary neural network-based methods. Section VI discusses the limitations of our approach. Finally, Section VII concludes this article.

II. RELATED WORK

The historical context of link prediction in complex networks is characterized by a variety of concepts [7], [24], [26] and even goes back to the early work of Jaccard [25]. Our models are built upon these concepts within the framework of dynamic, scale-free-like networks to identify anomalous links.

Previous studies on anomaly detection in scale-free networks employ node representations or embeddings. CenGCN [27] employs label propagation with labeled hub vertices to quantify the similarity between hub vertices and their neighbors. The proposed method captures the differentiation in information passing between vertices. Feng et al. [28] proposed a network embedding technique by implementing a “degree penalty” principle, which punishes the proximity between high-degree nodes.

Rahman et al. [29] developed a hybrid statistical approach to classify anomalous users in social networks. Liu et al. [30] introduced a Bayesian statistical model in combination with a time-slicing function to enhance the identification of novel and unfamiliar anomalous nodes. Gu et al. [18] extended a graph neural network to develop a network representation learning algorithm. The proposed algorithm encodes graphs into low-dimensional vector representations for node centrality measurement and community detection tasks. These studies employ node embeddings to detect anomalous nodes using only static graphs. As a result, these methods are unable to detect anomalous links and cannot capture the evolving communication patterns.

Few studies use probabilistic approaches to detect anomalies. Bars et al. [11] developed a probabilistic framework to detect abnormal communication volume at the node-level. The proposed framework infers the conditional probabilities of a “clique” generation and assigns an anomaly score to each node. Safdari and De Bacco [31] proposed a probabilistic generative approach to identify potential anomalies deviating from expected patterns. The model uses latent variables for community membership and anomaly parameters to detect anomalies on network edges. Park et al. [32] proposed a method to detect anomalies based on probabilistic analysis. The approach models correlated features within a multivariate normal distribution, such as TCP throughput and CPU usage. While these probabilistic approaches effectively detect anomalous nodes on static graphs, they are often limited by their reliance on predefined statistical assumptions, such as clique structures or multivariate normal distributions.

Several studies use dynamic graphs for anomalous node detection. Tian et al. [13] introduced an encoder-based node anomaly detector and use a time-equipped memory bank to record the overall statistical distributions of normal and unlabeled samples. Heard et al. [14] developed a two-stage Bayesian model to detect anomalous nodes in social networks. Ding et al. [12] proposed a deep autoencoder framework to reconstruct the original attributed network with the learned node embeddings from GCN and detect anomalous nodes. Although these methods are developed over dynamic graphs, they only detect anomalous nodes. This limits their applicability in scenarios where unexpected interactions or malicious connections between nodes are critical for anomaly detection.

Some studies detect anomalous links in dynamic graphs. SEDANSPOT [33] employs a random walk-based edge anomaly scoring function and exploits the evolving graph structures to detect anomalous edge streams. Bhatia et al. [34] developed an intrusion detection technique by combining statistical (chi-squared test) and algorithmic (count-min sketch) approaches to detect anomalous edge streams. Cai et al. [16] extracted h -hop subgraphs centered on the target edge to leverage unusual subgraph structures and employ a graph neural network to detect anomalous links. Xu et al. [15] proposed sampling subgraphs from the poisoned training data and employing outlier detection methods to detect suspicious adversarial edges. King and Huang [35] proposed an intrusion detection framework to detect anomalous traffic involving network lateral movement. Lo et al. [17] detected anomalous traffic in IoT networks. They utilize network traffic datasets where both normal and attack traffic are represented as labels in the network graph and do not consider the unobserved links. While these methods effectively

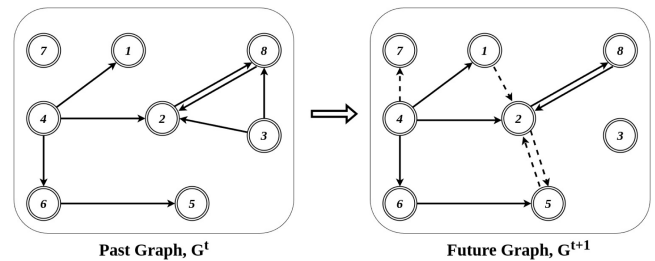


Fig. 1. Example of edges appearing and disappearing in a network between two time intervals; $[t - 1, t)$, $[t, t + 1)$.

detect anomalous links in dynamic graphs, they rely heavily on node-neighborhood or edge-based subgraphs, which are computationally expensive.

Other studies employ graph neural networks over directed graphs for link prediction task. Tong et al. [36] proposed a spectral-based graph convolutional network for directed graphs. The proposed neural network effectively learns directed graph representation by first- and second-order proximity matrices. Zhang et al. [37] proposed a spectral graph neural network for directed graphs. The authors incorporate the directional information via a complex Hermitian matrix. Gasteiger et al. [38] introduced directional message passing that enables graph neural networks to leverage direction information for molecular interaction predictions. Nevertheless, these studies do not detect anomalous links. As a result, they are less effective in identifying anomalous communication patterns.

Our study is different in two ways with respect to the existing approaches discussed above. First, our approach leverages the node degrees to compute future link probabilities without requiring additional information such as node features, node attributes, or edge features. Second, our developed model, Model-C, defined over directed graphs, can predict the direction of the links.

III. ANOMALOUS LINK PREDICTION

Given that MITRE’s network and other focused networks exhibit scale-free-like behavior, we take advantage of this property to estimate the future link probabilities based on past data. Employing these probabilities allows us to automate access control in a networked system by allowing the links with higher probabilities (expected) while rejecting the links with lower probabilities (anomalous).

Let G be a dynamic, simple graph representing a networked system changing in time. Let our current time be t such that the time interval between $t - 1$ and t refers to the “past” and the time interval between t and $t + 1$ refers to the “future.” Let $G^t = (V^t, E^t)$ be the “past graph” representing the snapshot of G between time $t - 1$ and t . The set of nodes or vertices, V^t , of graph G^t corresponds to the devices in the networked system between time $t - 1$ and t . The set of links or edges, E^t , of graph G^t represents the communication taking place between pairs of devices in the networked system between time $t - 1$ and t . Fig. 1 illustrates an example of dynamically evolving nature of the networks. In the figure, solid edges represent repeated connections, and dashed edges represent new connections forming in two consecutive time intervals. Also, connections between node pairs (3,2) and (3,8) disappear from time t to

$t + 1$. Our first goal in this study is to estimate the likelihood of a link (communication) between any pairs of nodes (networked devices) in the future, i.e., from time t to $t + 1$. To achieve this goal, we develop probabilistic models that use the past graph $G^t = (V^t, E^t)$ to assign a probability to any future graph configuration $G^{t+1} = (V^{t+1}, E^{t+1})$. Our second goal is to present an algorithm that consumes the estimated probabilities to decide whether the communication between two devices in the networked system is unexpected (anomalous), hence, to be blocked or expected (ordinary), hence, to be allowed.

One important feature of the models developed in this study is that any communication attempt involving a device that has never been observed in the past is flagged as anomalous. To put in other words, links (edges) involving unseen nodes (vertices) should assume zero probability. We believe that such communication attempts should be blocked by default until the device and behavior are vetted by a security officer or systems administrator. Our belief is also coherent with the security principle of “fail-safe permission default” [39], [40] meaning that the default access control configuration should conservatively protect a system. The principle emphasizes that systems should operate under the assumption that unknown entities could pose potential risks, thereby prioritizing security over convenience. In our models, this principle ensures that no implicit trust is granted to new or unknown devices, thereby enforcing the overall security of the network. This cautious approach is particularly critical in networks where malicious actors could exploit vulnerabilities by introducing new and unverified nodes. While this design enhances the security of the system, it may cause temporary disruptions for legitimate new devices until their behavior is classified. However, this tradeoff is necessary to prevent unauthorized access and maintain the integrity of the system. By requiring the intervention of a security officer or systems administrator, our models achieve a balance between security and usability, safeguarding the network while still supporting the addition of new devices.

A. Models for Anomalous Link Prediction

Preferential attachment index [7] of a link between two nodes is defined as the multiplication of the degrees of the two nodes. The index simply implies that the well-connected nodes in a graph are expected to have more connections. In the following, we first develop a probabilistic model based on the preferential attachment index. Then, we introduce modifications of the model to improve probability estimations in undirected and directed graphs. Finally, we present an algorithm that integrates any of the probabilistic models to detect anomalous links and automate or semiautomate access control decisions in a network domain.

1) *Model-A*: The first model assumes that the probability of an edge between two vertices is higher, when the degrees of the induced vertices are higher. The probability of an edge $e_{i,j}^{t+1} \in E^{t+1}$ between v_i^{t+1} and v_j^{t+1} is estimated by the following equation:

$$P(e_{i,j}^{t+1} \in E^{t+1}) = \frac{d_i^t d_j^t}{\sum_{k=1}^{|V^t|-1} \sum_{l=k+1}^{|V^t|} d_k^t d_l^t} \quad (1)$$

where d_i^t and d_j^t are the degrees of the vertices v_i^t and v_j^t in G^t . Note that the denominator of (1) is computed only once for all probability estimations.

The vertices which have never been seen in the past take degree zero. Therefore, any of their edges assume zero probability in (1). Coherent with the principle of “fail-safe permission default,” such edges are flagged as anomalous until vetted by a security officer or systems administrator and their vertices introduced into the system.

2) *Model-B*: The first model assigns higher probabilities to the links between two higher degree nodes compared to the links between a higher degree and a lower degree node. While this model is suitable for some scale-free networks, it does not necessarily fit the computer network domains. In a typical networked system, client nodes connect to various service nodes resulting in high-degree service nodes. However, often the service nodes are independent or partially dependent. Therefore, it is more suitable to treat the links between two higher degree nodes as well as a higher degree and a lower degree node similarly. The probability of an edge $e_{i,j}^{t+1} \in E^{t+1}$ between v_i^{t+1} and v_j^{t+1} is estimated by the following equation:

$$P(e_{i,j}^{t+1} \in E^{t+1}) = \frac{\max\{d_i^t, d_j^t\} \mathbb{1}_{V^{1:t}}(v_i^{t+1}) \mathbb{1}_{V^{1:t}}(v_j^{t+1})}{\sum_{k=1}^{|V^t|-1} (|V^t| - k) \pi_k^t} \quad (2)$$

where d_i^t, d_j^t are the degrees of the vertices v_i^t and v_j^t from the past graph, G^t , and π_k^t is the k th highest degree node in the past graph G^t . Indicator functions $\mathbb{1}_{V^{1:t}}(v_i^{t+1})$ and $\mathbb{1}_{V^{1:t}}(v_j^{t+1})$ ensure that both v_i^{t+1} and v_j^{t+1} have been seen in the past, starting from the first day. The indicator functions ensure that Model-B is coherent with the security principle of “fail-safe permission default,” i.e., (2) is zero for the edges involving unseen vertices. Note that the denominator of (2) is computed only once for all probability estimations.

3) *Model-C*: While Model-A and Model-B are defined over undirected graphs, the availability of direction information distinguishes between a link from one node to the other and its reverse. For example, communication data gathered at the transport layer of the TCP/IP protocol has a direction from client nodes, which “initiate” a connection, to service nodes. Note that a service node may also behave as a client node to receive some service from another node. Hence, Model-A and Model-B fail to capture the missions of the nodes in link prediction over data with direction information. When the direction information is available, the probability of an edge $e_{i,j}^{t+1} \in E^{t+1}$ from v_i^{t+1} to v_j^{t+1} is estimated by the following equation:

$$P(e_{i,j}^{t+1} \in E^{t+1}) = \frac{d_j^{\text{int}} \mathbb{1}_{V^{1:t}}(v_i^{t+1}) \mathbb{1}_{V^{1:t}}(v_j^{t+1})}{|E^t|(|V^t| - 1)} \quad (3)$$

where d_j^{int} is the in-degree of the vertex v_j^t from the past graph, G^t and $|E^t|$ is the edge counts, and $|V^t|$ is the vertex counts of the past graph, G^t . Indicator functions $\mathbb{1}_{V^{1:t}}(v_i^{t+1})$ and $\mathbb{1}_{V^{1:t}}(v_j^{t+1})$ ensure that both v_i^{t+1} and v_j^{t+1} have been seen in the past, starting from the first day. The indicator functions ensures that Model-C is coherent with the security principle of “fail-safe permission default,” i.e., (3) is zero for edges involving unseen vertices.

TABLE I
APPLICABILITY OF THE PROPOSED MODELS IN DIFFERENT
NETWORK SCENARIOS

Models	Network Type	Applicable Scenarios
Model-A	Undirected	Suitable for networks where communication is symmetric or directionality is irrelevant.
Model-B	Undirected	Useful in networks with partially dependent hub nodes.
Model-C	Directed	Best suited for networks where the direction of communication links is critical.

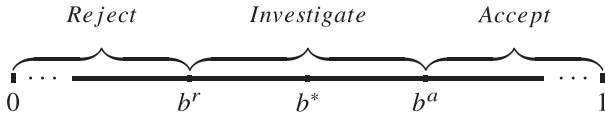


Fig. 2. Illustrative example of link decision boundaries.

Note that the denominator of (3) is computed only once for all probability estimations.

We summarize the applicability of the proposed models under different network conditions in Table I. Model A is best suited for undirected networks in which communication is symmetric, or the directionality of links is not critical. Model-B is particularly useful for undirected networks with partially dependent hub nodes. Model-C, on the other hand, is specifically designed for directed networks where the direction of communication links plays a crucial role.

B. Algorithm for Anomalous Link Decision

Since the computed probabilities are not calibrated, using 0.5 probability threshold as a decision boundary to automatically separate anomalous links from the expected ones does not work under any model. In this study, we use the ROC curve [41] of different threshold values to find the best decision boundary among many candidates. ROC curve plots true positive rate (TPR) against the false positive rate (FPR) at various boundary values. The optimal decision boundary is typically the one on the upper-left corner of the ROC curve. We use the optimization presented in (4) to find the best decision boundary, b^* among a sequence of candidate values, b

$$b^* = \arg \min_b |\text{TPR}(b) + \text{FPR}(b) - 1|. \quad (4)$$

The best decision threshold requires a grid search between 0 and 1. In practice, though, the search is conducted over an interval spanning from the minimum probability to the maximum probability reported by each model. The practical minimum and maximum probabilities are obtained while computing the denominators of (1)–(3).

Algorithm 1 requires a reference to the object representing one of the probabilistic models, M that was run on the past graph G^t . In addition to the model, the algorithm requires the link to be tested, $e_{i,j}^{t+1}$, the maximum threshold, b^r for an automatic reject, and the minimum threshold, b^a for an automatic accept. Algorithm 1 estimates the probability of the input link $e_{i,j}^{t+1}$ and fetches the optimal boundary at lines 1 and 2, respectively. As illustratively depicted in Fig. 2, the conditions at lines 3 and 4, and lines 5 and 6 in Algorithm 1, automatically reject or accept

Algorithm 1: Anomalous Link Decision.

Require: M $\triangleright$ the object reference for one of the probabilistic models
Require: $e_{i,j}^{t+1}$ $\triangleright$ the communication link to be tested
Require: b^r $\triangleright$ link rejection threshold
Require: b^a $\triangleright$ link acceptance threshold

- 1: $p \leftarrow M.Estimate.Probability(e_{i,j}^{t+1})$
- 2: $b^* \leftarrow M.Optimal.Boundary()$
- 3: **if** $p < b^r$ **then**
- 4: Automatically reject link $e_{i,j}^{t+1}$
- 5: **else if** $p \geq b^a$ **then**
- 6: Automatically accept link $e_{i,j}^{t+1}$
- 7: **else**
- 8: Investigate link $e_{i,j}^{t+1}$ further
- 9: **end if**

the tested communication link. Lines 7 and 8, on the other hand, require human intervention to investigate the test link.

In Algorithm 1, by default, the link decisions are made as follows:

- 1) a link is rejected if its probability is below b^r ;
- 2) a link is accepted if its probability is above or equal to b^a ;
- 3) a link requires further investigation by a security officer or a system administrator if its probability is between b^r and b^a .

Algorithm 1 looks like a simple algorithm, but one can enrich its behavior by controlling the threshold probabilities b^r and b^a . For example, when $b^r = b^a = b^*$ the system will be fully automated allowing any link with probability above or equal to b^* to be accepted and any link with probability below b^* to be rejected. The system will still be fully automated when $b^r = b^a < b^*$, but it will be more forgiving, i.e., lenient toward accepting links. Similarly, the system will still be fully automated when $b^r = b^a > b^*$, but it will be stricter this time, i.e., lenient toward rejecting links.

IV. EMPIRICAL EVALUATIONS

In this section, we describe the dataset, present the synthetic data generation process, and provide preprocessing details. Next, we examine the designations of the scale-free property of our dataset. Then, we outline the experimental design of the study. Finally, we present the empirical evaluation of the proposed models.

A. Description

Enterprises are naturally reluctant to share their network topology or traffic flow information because of security and privacy concerns. On the other hand, working with realistic datasets is essential to understand the behavior of enterprise networks. In this study, we use a communication network dataset provided by the MITRE Corporation. The dataset is synthetically generated using an AI model that processes a real-world communication network and produces synthetic versions, while retaining the authentic characteristics (including topological statistics, node features, and edge dynamics) of the original network flow data. The dataset spans over eight days and contains network flow information, including the source IP address, destination IP

address, starting and ending timestamps of the flows, destination port, and protocol.

B. Synthetic Dataset Generation Process

MITRE employs a probabilistic model to generate synthetic dynamic multigraphs that capture network structure, edge dynamics, and port-protocol combinations while preserving statistical properties of real-world network data. They introduce a diversity metric to assess the model's capability to generate a diverse set of graphs while maintaining similarity to a reference graph. The generative framework consists of three stages: *structure generation*, *feature generation*, and *graph alignment*. During the structure generation stage, a stochastic Kronecker graph generator [42] transforms the reference dynamic multigraph into a static graph. The KronFit algorithm [42] optimizes the Kronecker graph generator using maximum likelihood estimation to ensure that the generated graph closely matches the degree distribution, clustering coefficient, and connectivity patterns of the reference network. In the feature generation stage, a conditional generative adversarial network (CTGAN) [43] synthesizes edge attributes such as start time, duration, and port-protocol combinations, ensuring that the generated dataset retains the temporal and categorical distributions of real network flows. CTGAN is particularly effective for structured tabular data, as it applies normalization for continuous attributes and one-hot encoding for categorical variables, preserving realistic statistical patterns. The final stage, graph alignment, embeds the generated features into the graph structure and determines the source and destination of edges. This step employs an XGBoost model [44], leveraging node centrality measures (e.g., degree, betweenness, eigenvector centrality) and temporal edge activity to infer probable edge connections. Dividing the process into distinct stages allows for easy adaptation to graphs with different node and edge counts compared to the reference graph. This approach also ensures that the synthetic graphs achieve adequate similarity to the reference graphs while maintaining sufficient diversity to obscure the specifics of the original graphs.

C. Dataset Preprocessing

Networked systems exhibit predictable communication behavior from one-time interval to the next. The duration of time intervals depends on the application domain. Using 24-h or one-day time intervals is more suitable in our case, as this is the network of a branch of an enterprise with many employees. Hence, we divided the dataset into eight days based on the starting timestamps. We observed only four edges and eight nodes on Day₈. However, these connections appear right after midnight of Day₇. So, we ignore Day₈ and utilize the data in the first seven days. We present the day-wise statistics of the dataset in Table II. The number of edges represents multiple connections between two devices, while the number of unique edges represents distinct connections between two devices. Moreover, we found that roughly 500 nodes appear or disappear over distinct time intervals. Since these nodes are part of the original network, we add these nodes to our daily graphs as isolated nodes. Consequently, our network representation in each graph snapshot includes all nodes in the network over seven days.

TABLE II
DAY-WISE STATISTICS OF THE MITRE DATASET

Days	No. of Nodes	No. of Edges	No. of Unique Edges	Density	No. of Isolated Nodes
Day ₁	1822	32197	1810	0.01808	487
Day ₂	1822	31738	1811	0.01801	494
Day ₃	1822	29614	1817	0.01675	492
Day ₄	1822	28265	1818	0.01541	467
Day ₅	1822	31618	1815	0.01813	501
Day ₆	1822	29102	1816	0.01608	476
Day ₇	1822	30038	1811	0.01710	496

Hence, our models are capable of making predictions involving isolated nodes.

Next, we convert the daily network dataset into undirected and directed simple graphs for each day separately to obtain training and test datasets. We extract the edge list based on the observed or existing links between node pairs from the graphs. Note that the number of possible node pairs is in quadratic order with respect to the number of nodes in the graph. However, only a small portion of node pairs have edges between them. The density of daily graphs indicates that only 1.5%–1.8% out of all possible node pairs have links between them, while the remaining ~98% of the pairs do not have links. Existing links between node pairs are considered expected links, and nonexisting links between node pairs are considered anomalous links in training and test datasets. Furthermore, we include two types of anomalous links in our datasets, namely, two-way-missing links and one-way-missing links. A nonexisting link between a node pair in both directions is considered as a “two-way-missing link.” On the other hand, a nonexisting link between a node pair in the opposite direction of an existing link between the same node pair is considered as a “one-way-missing link.” As the number of expected and anomalous links is significantly imbalanced, we use random sampling on the anomalous links (including both two-way and one-way-missing links) to balance the datasets. Hence, each dataset contains about 50% expected, 25% two-way-missing anomalous, and 25% one-way-missing anomalous links. We set the label of expected links to 0 and anomalous links to 1. Finally, the dataset generated from Day_N is used as the training dataset for the test dataset generated from Day_{N+1}, where $N = \{1, 2, 3, 4, 5, 6\}$.

D. Scale-Free Property Test

We perform a statistical test developed in [8] to ensure that our dataset exhibits the scale-free property. There are six designations based on the scale-freeness of the networks: super-weak, weakest, weak, strong, strongest, and not scale-free [8]. We test whether the daily graphs satisfy one of the designations of scale-free property and present our findings in Table III. Day₂, Day₃, and Day₆ adhere to the “weak” designation at best, and the remaining four days adhere to the “weakest” designation. Therefore, we conclude that our daily datasets exhibit scale-free-like behavior.

E. Experimental Design

First, we compute the link probabilities on the training (Day₁) dataset using (1), (2), and (3) for Model-A, Model-B, and

TABLE III
SCALE-FREE PROPERTY OF THE DAILY GRAPHS OF THE MITRE DATASET

Daily Graphs	Strongest	Strong	Weak	Weakest	Super Weak	Not Scale-free
Day ₁	×	×	×	[✓]	[✓]	×
Day ₂	×	×	[✓]	[✓]	[✓]	×
Day ₃	×	×	[✓]	[✓]	[✓]	×
Day ₄	×	×	×	[✓]	[✓]	×
Day ₅	×	×	×	[✓]	[✓]	×
Day ₆	×	×	[✓]	[✓]	[✓]	×
Day ₇	×	×	×	[✓]	[✓]	×

TABLE IV
PERFORMANCE OF MODEL-A ON TEST (DAY₂) DATASET

Accuracy	Precision	Recall	F1 Score
63.46%	61.50%	71.93%	66.31%

TABLE V
CONFUSION MATRIX OF MODEL-A ON TEST (DAY₂) DATASET

Actual	Predicted	
	Expected	Anomalous
Expected	996	815
Anomalous	508	1302

Model-C, respectively. Then, we perform a 10-fold cross-validation to obtain the optimal decision threshold using (4). The training (Day₁) dataset is divided into train and validation sets. We use stratified k-fold cross validation [45] to balance expected and anomalous links in each fold. The best-performing threshold across 10-folds is selected as the optimal decision boundary. Next, we compute the link probabilities and apply the selected decision boundary to classify the anomalous links and expected links on the test (Day₂) dataset. Then, we present and discuss the classification results of the test (Day₂) dataset. We repeat the same process for all daily consecutive datasets, that is, we compute the link probabilities and obtain the optimal decision boundary from the training (Day_N) datasets. Then, we apply the decision boundary and classify the links from the test (Day_{N+1}) datasets. Next, we present the evaluation metrics of our models across all test datasets. Finally, we summarize the performance of our models based on anomalous link detection and expected link prediction tasks.

F. Empirical Results

In the following, we present and analyze the empirical results of each model individually.

1) *Performance of Model-A*: AUROC scores on the training (Day₁) dataset reach up to 74%. We found that the optimal decision threshold is 2.27×10^{-5} . Then, we apply the decision threshold on the test (Day₂) dataset. Table IV demonstrates that Model-A achieves an accuracy of 63.46%, precision of 61.50%, recall of 71.93%, and F1 score of 66.31%. Next, we present the confusion matrix for the test (Day₂) dataset in Table V to analyze the results. Model-A misclassifies almost half of the expected links as anomalous.

We investigated the empirical cumulative distribution function (ECDF) of node degrees to gain more insights. Our findings indicate that more than 73% of the nodes have very low degrees

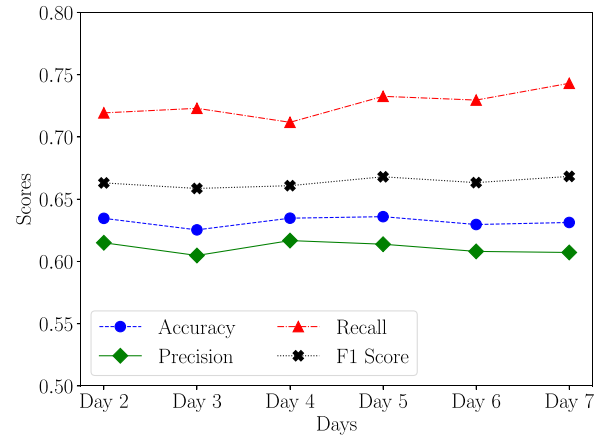


Fig. 3. Performance of Model-A across all daily test datasets.

TABLE VI
PERFORMANCE SUMMARY OF MODEL-A ON AVERAGE ACROSS ALL TEST DATASETS

Accuracy	Anomalous Link Detection			Expected Link Prediction		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
63.20%	61.10%	72.66%	66.37%	66.29%	53.74%	59.35%

TABLE VII
PERFORMANCE OF MODEL-B ON TEST (DAY₂) DATASET

Accuracy	Precision	Recall	F1 Score
66.50%	66.95%	65.20%	66.06%

(less than 3). Consequently, Model-A estimates a lower probability to the majority of expected links connecting to two lower degree nodes and misclassifies them as anomalous. Next, we present day-wise evaluation metrics in Fig. 3. Model-A maintains consistent performance across all remaining test datasets, where the difference is between two to four points. Finally, we summarize the performance of Model-A across all test datasets in Table VI. Model-A obtains an accuracy of 63.20% on average across all test datasets. In addition to detecting anomalous links, our proposed model demonstrates the capability to predict future expected links. Hence, we also present the results of the expected link prediction in Table VI. Model-A achieves an F1 score of 66.37% for anomalous link detection and 59.35% for expected link prediction tasks. Note that anomalous link detection is our primary goal, while the expected link prediction is a secondary outcome of our approach.

2) *Performance of Model-B*: AUROC scores of Model-B also reaches up to 74% on the training (Day₁) dataset. We found that the optimal decision threshold is 3.32×10^{-5} . Table VII demonstrates the results of the test (Day₂) dataset after applying the decision threshold. Model-B achieves an accuracy of 66.5%, precision of 66.95%, recall of 65.20%, and F1 score of 66.06%. Next, we present the confusion matrix in Table VIII. We observe that Model-B improves over Model-A in predicting expected links. This can be attributed to Model-B's ability to treat the links between two higher degree nodes similarly to a higher degree and a lower degree node. Next, we present day-wise evaluation metrics in Fig. 4. Model-B maintains consistent performance

TABLE VIII
CONFUSION MATRIX OF MODEL-B ON TEST (DAY₂) DATASET

Actual	Predicted	
	Expected	Anomalous
Expected	1238	573
Anomalous	620	1190

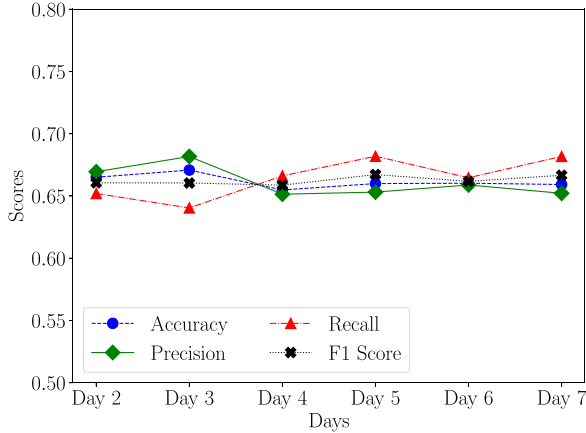


Fig. 4. Performance of Model-B across all daily test datasets.

TABLE IX
PERFORMANCE SUMMARY OF MODEL-B ON AVERAGE ACROSS ALL TEST DATASETS

Accuracy	Anomalous Link Detection			Expected Link Prediction		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
66.17%	66.11%	66.45%	66.25%	66.27%	65.89%	66.15

TABLE X
PERFORMANCE OF MODEL-C ON TEST (DAY₂) DATASET

Accuracy	Precision	Recall	F1 Score
96.85%	96.54%	97.18%	96.86%

across all remaining test datasets, varying between two to five points only. Lastly, we summarize the performance of Model-B across all test datasets in Table IX. Model-B achieves accuracy scores of 66.17% on average across all test datasets. Moreover, the F1 scores for anomalous link detection and expected link prediction suggest that Model-B performs similarly for both tasks. Furthermore, Model-B outperforms Model-A with a three-point improvement in accuracy scores.

3) *Performance of Model-C*: Model-A and Model-B cannot capture the direction of the links as they are developed over undirected graphs. Model-C is developed to take advantage of the direction information. Model-C achieves above 99% AU-ROC score. We found that the optimal decision threshold is 9.1×10^{-7} . Then, we present the results from the test (Day₂) dataset in Table X. Model-C obtains a remarkable accuracy of 96.85%, precision of 96.54%, recall of 97.18%, and F1 score of 96.86%. The confusion matrix presented in Table XI shows that Model-C adeptly identifies both expected and anomalous links. Moreover, the number of misclassified samples has reduced significantly compared to other models. We investigated the ECDF of node degrees over directed graphs and found that

TABLE XI
CONFUSION MATRIX OF MODEL-C ON TEST (DAY₂) DATASET

Actual	Predicted	
	Expected	Anomalous
Expected	1748	63
Anomalous	51	1759

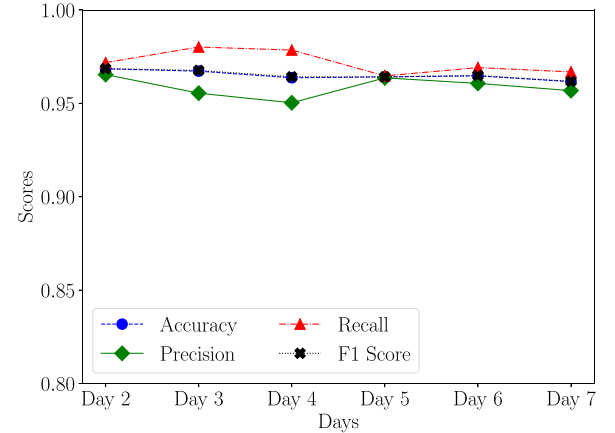


Fig. 5. Performance of Model-C across all daily test datasets.

TABLE XII
PERFORMANCE SUMMARY OF MODEL-C ON AVERAGE ACROSS ALL TEST DATASETS

Accuracy	Anomalous Link Detection			Expected Link Prediction		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
96.50%	95.87%	97.19%	96.52%	97.16%	95.81%	96.48%

more than 94% of nodes have zero in-degrees based on the test (Day₂) dataset. Only 97 out of 1822 nodes receive connections from other nodes. Therefore, only a few nodes have very high in-degrees, while the majority of the nodes have very low or zero in-degrees. Model-C takes advantage of this fact to accurately classify anomalous and expected links. Next, we present day-wise evaluation metrics in Fig. 5. Accuracy, recall, and F1 scores are consistently more than 96%, while precision scores are over 95%. Lastly, we summarize the performance of Model-C across all datasets in Table XII. Model-C obtains accuracy scores of 96.50% on average across all test datasets, while the other two models achieve 63.20% and 66.17%, respectively. F1 scores for anomalous link detection and expected link prediction are also similar. Furthermore, Model-C outperforms other models with at least a 30-point improvement in accuracy scores.

G. Worst-Case Scenario

In client-server models, the majority of the client nodes typically connect to a limited number of server nodes, resulting in server nodes exhibiting high in-degrees. In contrast, clients tend to have lower or even zero in-degrees. However, when a server node is compromised, it may initiate random and unusual connections to client nodes, which have previously received services. These links appear in the opposite direction of previously observed links. Such connections are considered anomalous links and are represented in our study as one-way-missing links.

TABLE XIII
PERFORMANCE SUMMARY OF THE PROPOSED MODELS ON AVERAGE ACROSS
ALL WORST-CASE DATASETS

Models	Accuracy	Precision	Recall	F1 Score
Model-A	47.41%	100.00%	47.41%	64.31%
Model-B	34.15%	100.00%	34.15%	50.88%
Model-C	96.66%	100.00%	96.66%	98.30%

Note that precision scores reach 100%, and recall exhibits identical scores as accuracy because these datasets consist of only one class.
The bold values are the best values according to the results.

Determining the direction of such links is crucial for effectively detecting the anomaly involving these connection attempts. We develop a *worst-case scenario*, where all the links appearing in the future are one-way-missing anomalous links. To that end, we create a separate test dataset, namely *worst-case dataset*, consisting of only one-way-missing links for each graph snapshot to test the robustness of our proposed models.

1) *Performance on Worst-Case Scenario*: Model-A and Model-B are defined over undirected graphs. Therefore, these models cannot predict the direction of the links. On the other hand, Model-C leverages the direction information. The model can detect one-way-missing links and predict the directionality of the links in the future graph. To support our claim, we test the trained models on the worst-case datasets. We summarize the performance of our proposed models for the worst-case scenario in Table XIII. Model-A and Model-B only achieve average accuracy scores of 47.41% and 34.15%, respectively, across all days. The results demonstrate that these models cannot detect the one-way-missing links due to the absence of direction information. On the contrary, Model-C has a superior performance compared to Model-A and Model-B on the worst-case scenario. The model achieves average accuracy scores of 96.66%. The ECDF of node degrees over directed graphs indicate that more than 94% of the nodes have zero in-degrees. As a result, the one-way-missing links possess zero or very low probability. Therefore, Model-C can accurately detect them as anomalous links by leveraging the direction information.

V. COMPARATIVE ANALYSES

In this section, we compare our proposed approach with two popular walk-based methods, i.e., Node2Vec [19] and DeepWalk [20], and two contemporary neural network-based methods, i.e., TADDY [21] and DGCN [22]. First, we present a brief overview of the comparison methods. Then, we provide the details of their experimental settings. Lastly, we discuss the performance of the walk-based and neural network-based methods against our proposed approach.

A. Overview of Comparison Methods

Node2Vec [19] combines the breadth-first search and depth-first search traversal algorithms to generate random walks. Node2Vec exploits the Skip-gram [46] technique to learn node representations by treating the sequences of nodes generated from the random walks as “sentences” and the nodes themselves as “words.”

DeepWalk [20] also utilizes a random-walk strategy to learn latent representations of nodes. It generates truncated random

walks with a given length starting from a target node. The method also exploits the Skip-gram [46] technique to generate node embeddings from random walks.

TADDY [21] is transformer-based model for detecting anomalies in dynamic graphs. It captures spatial and temporal patterns in evolving graphs using a novel spatial-temporal node encoding mechanism.

DGCN [22] uses GCNs to learn node embeddings in dynamic graphs while incorporating temporal information to ensure smooth representation evolution. By integrating graph structure and temporal dynamics, it effectively captures changes in relationships across graph snapshots.

B. Experimental Settings of Comparison Methods

To compare Node2Vec, DeepWalk, TADDY, and DGCN with our proposed approach, we adopted the best experimental settings as specified in their respective studies [19], [20], [21], [22]. These settings include parameters such as the number of random walks, walks per node, walk length, context size, number of workers, number of convolutional layers, and number of attention heads. In addition, we performed a grid search to fine-tune the return and in-out parameters for Node2Vec, as well as the learning rate and embedding dimensions for all methods. We generated node embeddings from the training datasets and applied the Hadamard operator to obtain edge embeddings for both training and test datasets.

C. Comparative Evaluation

Fig. 6 presents the comparative evaluation of the walk-based and neural network-based methods with our proposed models over undirected graphs. The accuracy, precision, and recall scores in Fig. 6 depict that all methods maintain consistent performance across daily test datasets, varying only by two to five points. Node2Vec and DeepWalk outperform Model-A by nine to thirteen points and Model-B by six to eleven points in accuracy scores. Similarly, TADDY and DGCN outperform Model-A by eleven to fifteen points and Model-B by nine to fourteen points. Next, we present the edge embeddings learned from Node2Vec, DeepWalk, TADDY, and DGCN in Fig. 7. We reduce the dimension of the edge embeddings to two using t-SNE [47]. Node2Vec and DeepWalk rely on random walks and the Skip-gram technique for learning node embeddings, resulting in similar patterns in the embedding space as depicted in Fig. 7(a) and (b). We also observe that nearly half of the expected links form a cluster, while the remaining expected links are scattered, with some inseparable from anomalous links. Consequently, Node2Vec and DeepWalk reach up to 78% and 74% in accuracy scores, respectively. In contrast, neural network-based methods, TADDY and DGCN, demonstrate slightly better separation between anomalous and expected links, as shown in Fig. 7(c) and (d). Thus, TADDY and DGCN achieve accuracy scores of 80% and 78%, respectively. Finally, we summarize the performance of all methods over undirected graphs across anomalous link detection and expected link prediction tasks in Table XIV. All methods outperform Model-A and Model-B on both tasks across all evaluation metrics. Both Model-A and Model-B misclassify the expected links connecting to two lower degree nodes as anomalous. On the other hand, the walk-based

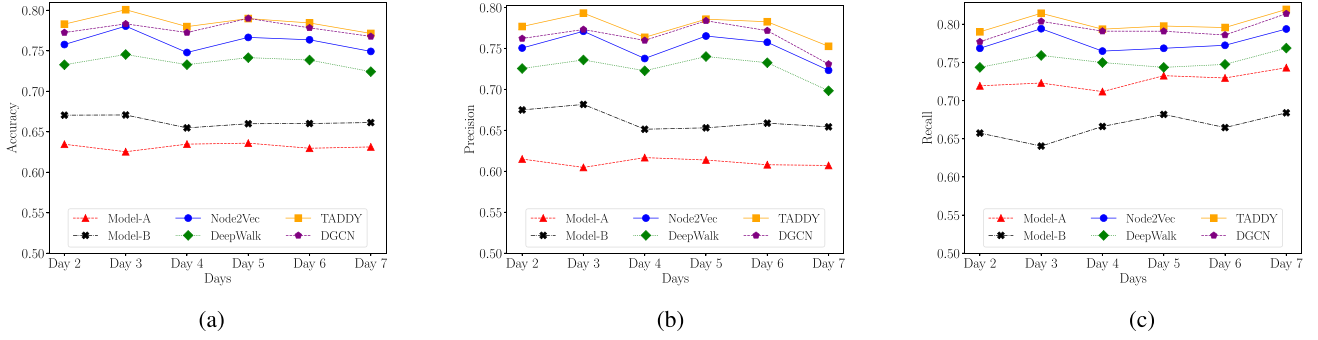


Fig. 6. Performance comparison of two popular walk-based methods and two contemporary neural network-based methods with Model-A and Model-B, over undirected graphs across all daily test datasets. (a) Accuracy. (b) Precision. (c) Recall.

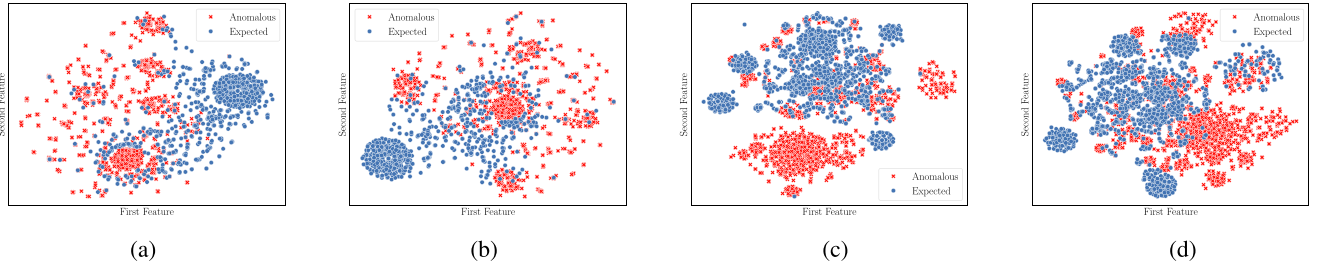


Fig. 7. Edge embeddings visualization using t-SNE, learned by Node2Vec, DeepWalk, TADDY, and DGCN over undirected graph on test (Day₂) dataset. (a) Node2Vec. (b) DeepWalk. (c) TADDY. (d) DGCN.

TABLE XIV

PERFORMANCE COMPARISON OF TWO POPULAR WALK-BASED METHODS AND TWO CONTEMPORARY NEURAL NETWORK-BASED METHODS WITH MODEL-A AND MODEL-B OVER UNDIRECTED GRAPHS ON AVERAGE ACROSS ALL TEST DATASETS

Methods	Accuracy	Anomalous Link Detection			Expected Link Prediction		
		Precision	Recall	F1 Score	Precision	Recall	F1 Score
Node2Vec	76.09%	75.09%	77.70%	76.36%	77.21%	74.49%	74.89%
DeepWalk	73.59%	72.59%	75.20%	73.86%	74.71%	71.99%	72.39%
TADDY	78.49%	77.58%	80.18%	79.08%	79.81%	76.90%	77.28%
DGCN	77.74%	76.36%	79.37%	77.88%	78.69%	75.60%	77.05%
Model-A	63.20%	61.10%	72.66%	66.37%	66.29%	53.74%	59.35%
Model-B	66.30%	66.24%	66.58%	66.38%	66.40%	66.02%	66.19%

The bold values are the best values according to the results.

TABLE XV

PERFORMANCE COMPARISON OF TWO POPULAR WALK-BASED METHODS AND TWO CONTEMPORARY NEURAL NETWORK-BASED METHODS WITH MODEL-C OVER DIRECTED GRAPHS ON AVERAGE ACROSS ALL TEST DATASETS

Methods	Accuracy	Anomalous Link Detection			Expected Link Prediction		
		Precision	Recall	F1 Score	Precision	Recall	F1 Score
Node2Vec	86.12%	85.46%	87.08%	86.25%	86.82%	85.16%	85.97%
DeepWalk	83.39%	82.41%	85.32%	83.88%	84.63%	81.87%	83.23%
TADDY	89.92%	89.56%	91.20%	90.27%	90.95%	89.44%	89.88%
DGCN	87.66%	86.87%	89.40%	87.96%	88.56%	86.11%	87.66%
Model-C	96.50%	95.87%	97.19%	96.52%	97.16%	95.81%	96.48%

The bold values are the best values according to the results.

and neural network-based methods do not solely rely on node degrees. Hence, they achieve better performance than the proposed models over undirected graphs.

Fig. 8 presents the comparative evaluation of the walk-based and neural network-based methods with our proposed model, Model-C, over directed graphs. In the figure, their performance varies between three to seven points only, suggesting these methods maintain consistent performance on all daily test datasets. Based on accuracy scores, Model-C outperforms Node2Vec by nine to twelve points, DeepWalk by eleven to sixteen points, TADDY by three to eight points, and DGCN by five to eleven points. Next, we present the t-SNE plot of edge embeddings learned by the walk-based and neural network-based methods in Fig. 9. Although the expected and anomalous links have some overlap, they are better separated than undirected graphs for all methods. As a result, all methods achieve an improvement of around nine to eleven points compared to their performance on undirected graphs. Finally, we summarize the performance

of all methods over directed graphs across all test datasets for anomalous link detection and expected link prediction tasks in Table XV. Based on all evaluation metrics for both tasks, TADDY outperforms DGCN by two points, Node2Vec by three points, and DeepWalk by six points. However, Model-C outperforms both walk-based and neural network-based methods with at least a seven-point improvement. Model-C leverages the direction information of the links more effectively than the other methods to detect anomalous links and predict expected links.

We also compare the walk-based and neural network-based methods with our approach on the worst-case scenario. Since the direction of the links is vital to detect the one-way-missing links, we used the trained models of these methods over directed graphs to ensure a fair comparison. Table XVI summarizes the performance comparison of all methods with Model-C on the worst-case scenario. Node2Vec reaches up to 82.18%, DeepWalk reaches up to 80.73%, TADDY reaches up to 87.67%, and DGCN reaches up to 86.01% in accuracy scores. On the other hand, Model-C outperforms walk-based methods by at least

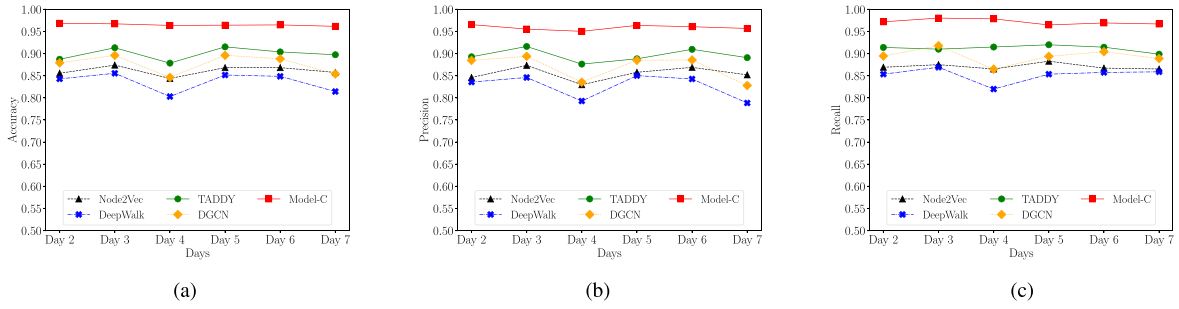


Fig. 8. Performance comparison of two popular walk-based methods and two contemporary neural network-based methods with Model-C over directed graphs across all daily test datasets. (a) Accuracy. (b) Precision. (c) Recall.

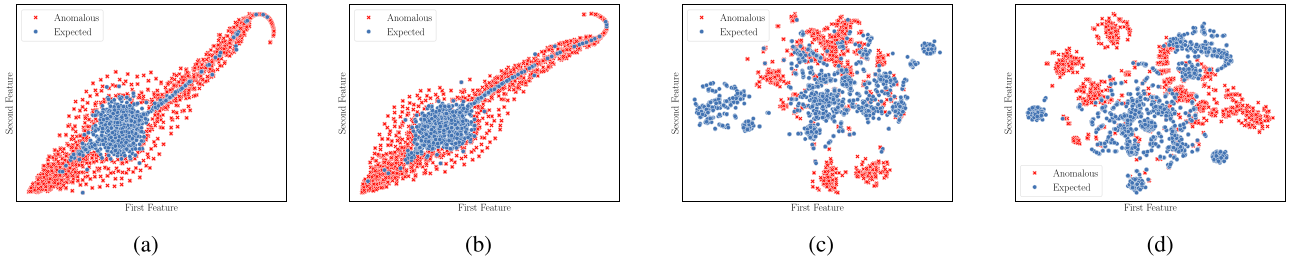


Fig. 9. Edge embeddings visualization using t-SNE, learned by Node2Vec, DeepWalk, TADDY, and DGCN over directed graph on test (Day₂) dataset. (a) Node2Vec. (b) DeepWalk. (c) TADDY. (d) DGCN.

TABLE XVI

PERFORMANCE COMPARISON OF TWO POPULAR WALK-BASED METHODS AND TWO CONTEMPORARY NEURAL NETWORK-BASED METHODS WITH MODEL-C ON AVERAGE ACROSS ALL WORST-CASE DATASETS

Methods	Accuracy	Precision	Recall	F1 Score
Node2Vec	82.18%	100.00%	82.18%	90.21%
DeepWalk	80.73%	100.00%	80.73%	89.33%
TADDY	87.67%	100.00%	87.67%	93.78%
DGCN	86.01%	100.00%	86.01%	92.18%
Model-C	96.66%	100.00%	96.66%	98.30%

The bold values are the best values according to the results.

14 points and neural network-based methods by at least nine points.

VI. LIMITATIONS

While the proposed models are effective in detecting anomalous links, they have certain limitations. One limitation is that the behavior of a new node or a previously unseen IP is not immediately known. As a result, its communication will be blocked by default until it is vetted by a systems administrator. Similarly, if the communication behavior of a node changes due to the user being assigned to a new or different project, the system will flag any communication attempt involving such node as anomalous. Another limitation arises when the IP addresses of nodes are translated (IP masquerading) by network address translation (NATs). In this case, the proposed approach will be unable to distinguish the behaviors of individual nodes behind the NAT. However, this limitation is not inherent to the approach itself but

is instead the result of deploying two conflicting systems on the same network: NAT and IP address-based profiling.

VII. CONCLUSION

In this study, we proposed three probabilistic models to detect anomalous links in dynamically evolving scale-free-like networks. Our proposed models leverage the scale-free-like property of the networked systems. We evaluated the proposed models on a dataset provided by the MITRE corporation that mimics a real-world communication network. The dataset contains network flows of seven days. We generated a graph snapshot for each day and converted the daily graph snapshots into undirected and directed simple graphs. We introduced a distinctive category of nonexistent links between node pairs, i.e., one-way-missing links, indicating connections that appear in the opposite direction of the existing links between the same node pairs. We included such one-way-missing links in the daily training and test datasets. Then, we developed three separate probabilistic models, namely Model-A, Model-B, and Model-C, defined over undirected and directed graphs. The empirical evaluations show that Model-C achieves an average accuracy of 96.50% across all test datasets, outperforming Model-A and Model-B by 33 and 30 points based on accuracy scores, respectively. Moreover, we showed that Model-C can successfully discern future link orientations. Furthermore, we developed a *worst-case scenario* consisting of only one-way-missing anomalous links to evaluate the robustness of our proposed models. We found that Model-C, defined over directed graphs, can accurately detect such one-way-missing links with an average accuracy of 96.66%. Lastly, we compared our approach with two popular walk-based and two

contemporary neural-network based methods, and demonstrated that Model-C outperforms across all test datasets.

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REFERENCES

- [1] M. S. M. Gismalla et al., "Survey on device to device (D2D) communication for 5G/6G networks: Concept, applications, challenges, and future directions," *IEEE Access*, vol. 10, pp. 30792–30821, 2022.
- [2] J. Logeshwaran, N. Shanmugasundaram, and J. Lloret, "Energy-efficient resource allocation model for device-to-device communication in 5G wireless personal area networks," *Int. J. Commun. Syst.*, vol. 36, no. 13, 2023, Art. no. e5524.
- [3] F. Al-Turjman, M. H. Nawaz, and U. D. Ulsar, "Intelligence in the Internet of Medical Things era: A systematic review of current and future trends," *Comput. Commun.*, vol. 150, pp. 644–660, 2020.
- [4] K. Pahlavan and P. Krishnamurthy, "Evolution and impact of Wi-Fi technology and applications: A historical perspective," *Int. J. Wireless Inf. Netw.*, vol. 28, pp. 3–19, 2021.
- [5] C. Lamba, "Security for high-speed MANs/WANs," in *Proc. 3rd Int. Conf. Recent Trends Netw. Secur. Appl.*, 2010, pp. 75–82.
- [6] J. Wang et al., "IoT-Praetor: Undesired behaviors detection for IoT devices," *IEEE Internet Things J.*, vol. 8, no. 2, pp. 927–940, Jan. 2021.
- [7] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, 1999.
- [8] A. D. Broido and A. Clauset, "Scale-free networks are rare," *Nature Commun.*, vol. 10, no. 1, 2019, Art. no. 1017.
- [9] P. Holme, "Rare and everywhere: Perspectives on scale-free networks," *Nature Commun.*, vol. 10, no. 1, 2019, Art. no. 1016.
- [10] H. Fereidooni, A. Dmitrienko, P. Rieger, M. Miettinen, A.-R. Sadeghi, and F. Madlener, "FedCRI: Federated mobile cyber-risk intelligence," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2022.
- [11] B. L. Bars and A. Kalogeratos, "A probabilistic framework to node-level anomaly detection in communication networks," in *Proc. IEEE Conf. Comput. Commun.*, 2019, pp. 2188–2196.
- [12] K. Ding, J. Li, R. Bhanushali, and H. Liu, "Deep anomaly detection on attributed networks," in *Proc. SIAM Int. Conf. Data Mining*, 2019, pp. 594–602.
- [13] S. Tian et al., "SAD: Semi-supervised anomaly detection on dynamic graphs," 2023, *arXiv:2305.13573*.
- [14] N. A. Heard, D. J. Weston, K. Platanioti, and D. J. Hand, "Bayesian anomaly detection methods for social networks," *Ann. Appl. Statist.*, vol. 4, no. 2, pp. 645–662, 2010.
- [15] X. Xu, H. Wang, A. Lal, C. A. Gunter, and B. Li, "EDoG: Adversarial edge detection for graph neural networks," in *Proc. IEEE Conf. Secure Trustworthy Mach. Learn.*, 2023, pp. 291–305.
- [16] L. Cai et al., "Structural temporal graph neural networks for anomaly detection in dynamic graphs," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manage.*, 2021, pp. 3747–3756.
- [17] W. W. Lo, S. Layeghy, M. Sarhan, M. Gallagher, and M. Portmann, "E-GraphSAGE: A graph neural network based intrusion detection system for IoT," in *Proc. IEEE/IFIP Netw. Operations Manage. Symp.*, 2022, pp. 1–9.
- [18] W. Gu, F. Gao, R. Li, and J. Zhang, "Learning universal network representation via link prediction by graph convolutional neural network," *J. Social Comput.*, vol. 2, no. 1, pp. 43–51, 2021.
- [19] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 855–864.
- [20] B. Perozzi, R. Al-Rfou, and S. Skiena, "DeepWalk: Online learning of social representations," in *Proc. 20th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2014, pp. 701–710.
- [21] Y. Liu et al., "Anomaly detection in dynamic graphs via transformer," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 12, pp. 12081–12094, Dec. 2023.
- [22] C. Gao, J. Zhu, F. Zhang, Z. Wang, and X. Li, "A novel representation learning for dynamic graphs based on graph convolutional networks," *IEEE Trans. Cybern.*, vol. 53, no. 6, pp. 3599–3612, Jun. 2023.
- [23] M. Hassan, M. E. Tozal, V. Swarup, S. Noel, R. Gottumukkala, and V. Raghavan, "Anomalous link detection in dynamically evolving scale-free-like networked systems," in *Proc. IEEE Int. Syst. Conf.*, 2024, pp. 1–8.
- [24] D. Liben-Nowell and J. Kleinberg, "The link prediction problem for social networks," in *Proc. 12th Int. Conf. Inf. Knowl. Manage.*, 2003, pp. 556–559.
- [25] P. Jaccard, "Étude comparative de la distribution florale dans une portion des alpes et des jura," *Bull. Soc. Vaudoise Sci. Nat.*, vol. 37, pp. 547–579, 1901.
- [26] M. E. Newman, "Clustering and preferential attachment in growing networks," *Phys. Rev. E*, vol. 64, no. 2, 2001, Art. no. 025102.
- [27] F. Xia et al., "CenGCN: Centralized convolutional networks with vertex imbalance for scale-free graphs," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 5, pp. 4555–4569, May 2023.
- [28] R. Feng, Y. Yang, W. Hu, F. Wu, and Y. Zhang, "Representation learning for scale-free networks," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1.
- [29] M. S. Rahman, S. Halder, M. A. Uddin, and U. K. Acharjee, "An efficient hybrid system for anomaly detection in social networks," *Cybersecurity*, vol. 4, no. 1, pp. 1–11, 2021.
- [30] T. Liu, A. Qi, Y. Hou, and X. Chang, "Method for network anomaly detection based on Bayesian statistical model with time slicing," in *Proc. IEEE 7th World Congr. Intell. Control Autom.*, 2008, pp. 3359–3362.
- [31] H. Safdari and C. De Bacco, "Anomaly detection and community detection in networks," *J. Big Data*, vol. 9, no. 1, 2022, Art. no. 122.
- [32] J. Park, D. H. Choi, Y.-B. Jeon, Y. Nam, M. Hong, and D.-S. Park, "Network anomaly detection based on probabilistic analysis," *Soft Comput.*, vol. 22, pp. 6621–6627, 2018.
- [33] D. Eswaran and C. Faloutsos, "SedanSpot: Detecting anomalies in edge streams," in *Proc. IEEE Int. Conf. Data Mining*, 2018, pp. 953–958.
- [34] S. Bhatia, R. Liu, B. Hooi, M. Yoon, K. Shin, and C. Faloutsos, "Real-time anomaly detection in edge streams," *ACM Trans. Knowl. Discov. Data*, vol. 16, no. 4, pp. 1–22, 2022.
- [35] I. J. King and H. H. Huang, "Euler: Detecting network lateral movement via scalable temporal link prediction," *ACM Trans. Privacy Secur.*, vol. 26, no. 3, pp. 1–36, 2023.
- [36] Z. Tong, Y. Liang, C. Sun, D. S. Rosenblum, and A. Lim, "Directed graph convolutional network," 2020, *arXiv:2004.13970*.
- [37] X. Zhang, Y. He, N. Brugnone, M. Perlmutter, and M. Hirn, "Magnet: A neural network for directed graphs," *Adv. neural Inf. Process. Syst.*, vol. 34, pp. 27003–27015, 2021.
- [38] J. Gastegger, J. Groß, and S. Günnemann, "Directional message passing for molecular graphs," 2020, *arXiv:2003.03123*.
- [39] J. H. Saltzer and M. D. Schroeder, "The protection of information in computer systems," *Proc. IEEE*, vol. 63, no. 9, pp. 1278–1308, Sep. 1975.
- [40] M. T. Goodrich and R. Tamassia, *Introduction to Computer Security*. London, U.K.: Pearson, 2011.
- [41] J. A. Hanley and B. J. McNeil, "The meaning and use of the area under a receiver operating characteristic (ROC) curve," *Radiol.*, vol. 143, no. 1, pp. 29–36, 1982.
- [42] J. Leskovec, D. Chakrabarti, J. Kleinberg, C. Faloutsos, and Z. Ghahramani, "Kronecker graphs: An approach to modeling networks," *J. Mach. Learn. Res.*, vol. 11, no. 2, 2010.
- [43] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [44] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 785–794.
- [45] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning*, vol. 112. Berlin, Germany: Springer, 2013.
- [46] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," *Adv. Neural Inf. Process. Syst.*, vol. 26, 2013.
- [47] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, 2008.